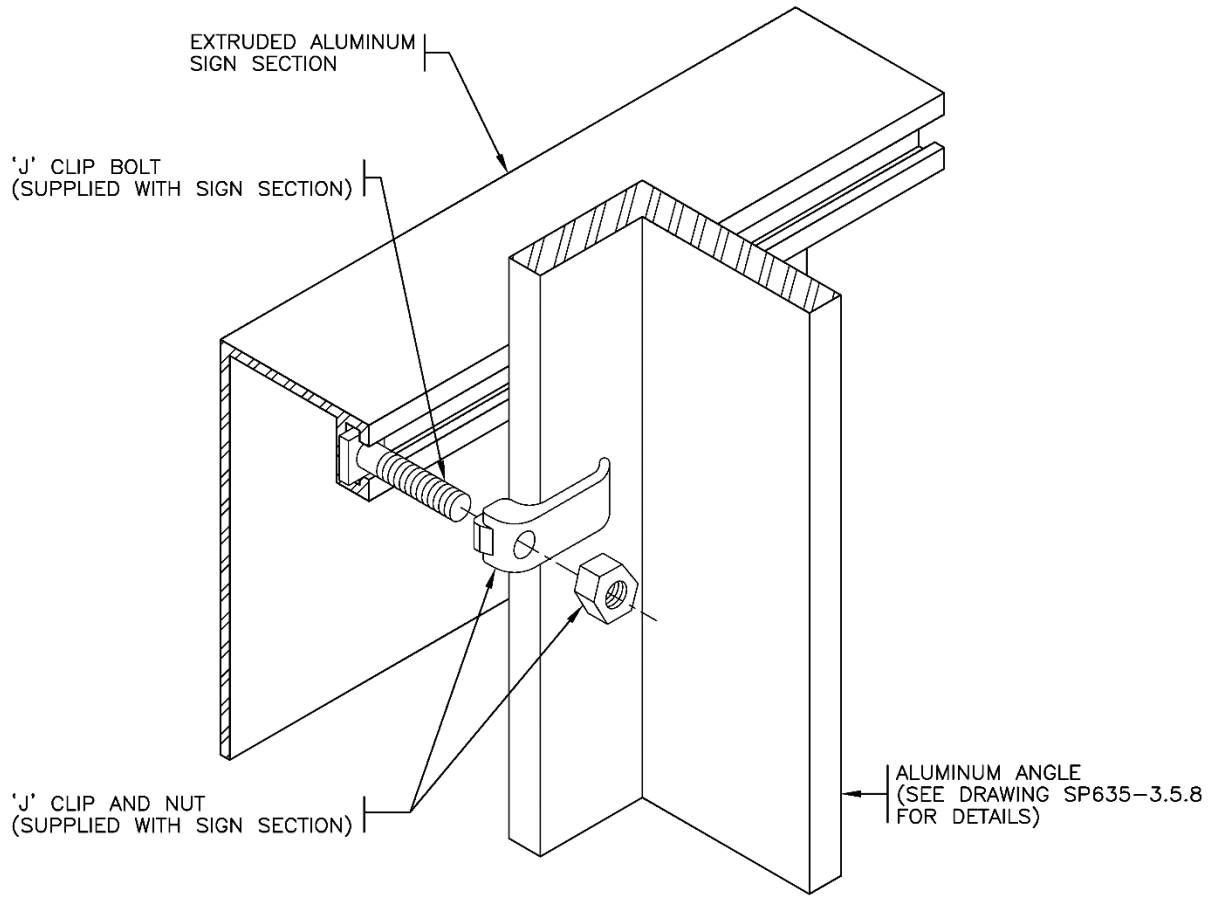


ASSEMBLY VIEW

TORQUE JOINER BOLTS AND 'J' CLIPS TO 19 ft. lbs.

SEE DRAWING SP635-3.5.6 FOR NOTES AND ADDITIONAL DETAILS.

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B		
A	GENERAL REVISIONS	APR 17

EXTRUDED ALUMINUM SIGN INSTALLATION DETAILS ON WOOD POST

Date Approved

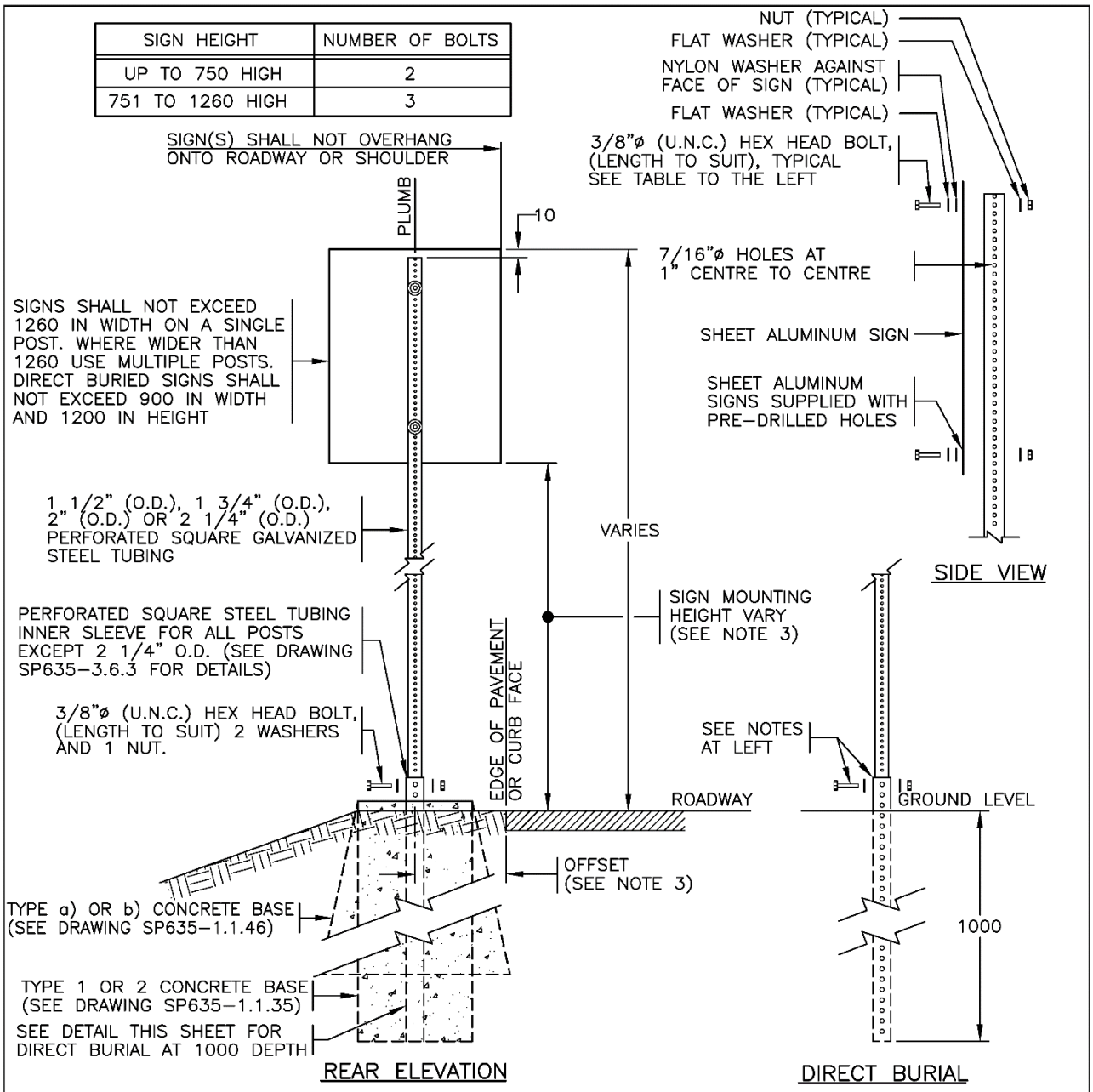
18/11/94 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-3.5.9

SIGN HEIGHT	NUMBER OF BOLTS
UP TO 750 HIGH	2
751 TO 1260 HIGH	3



REFER TO MINISTRY ELECTRICAL & TRAFFIC ENGINEERING MANUAL FOR SIGN STRUCTURE LOADING TABLE

ALL NUTS, BOLTS AND WASHERS SHALL BE STAINLESS STEEL

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. MOUNTING HEIGHTS & OFFSETS SHALL FALL WITHIN THE RANGES NOTED IN THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS & PAVEMENT MARKINGS.

NOT TO SCALE

BRITISH COLUMBIA — Ministry of Transportation and Infrastructure

No.	Revision	Date
G	GENERAL REVISIONS	APR 20
F	GENERAL REVISIONS	APR 17
E	TRAPEZOIDAL BASE & DIRECT BURIAL DETAIL ADDED	APR 05
D	MOUNTING HEIGHTS & OFFSETS REFERED THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS	NOV 98
C	SIGN MOUNTING HEIGHT REVISED	DEC 96

PERFORATED SQUARE GALVANIZED STEEL SIGN POST INSTALLATION DETAIL (ONE POST)

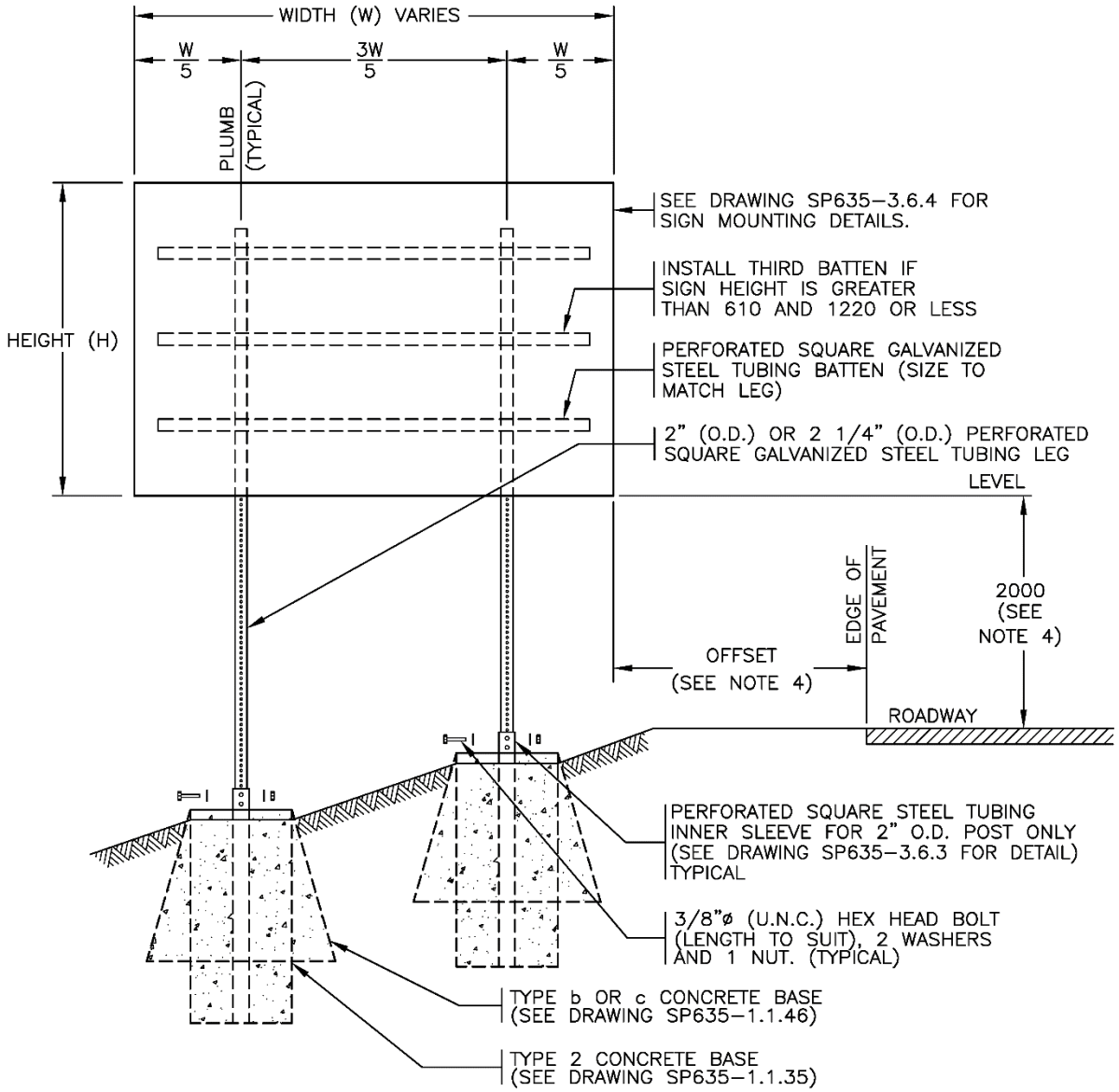
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-3.6.1



ELEVATION

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. ALL BOLTS, NUTS AND WASHERS SHALL BE STAINLESS STEEL.
4. MOUNTING HEIGHTS & OFFSETS SHALL FALL WITHIN THE RANGES NOTED IN THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS & PAVEMENT MARKINGS.

REFER TO MINISTRY ELECTRICAL & TRAFFIC ENGINEERING MANUAL FOR SIGN STRUCTURE LOADING TABLE

NOT TO SCALE

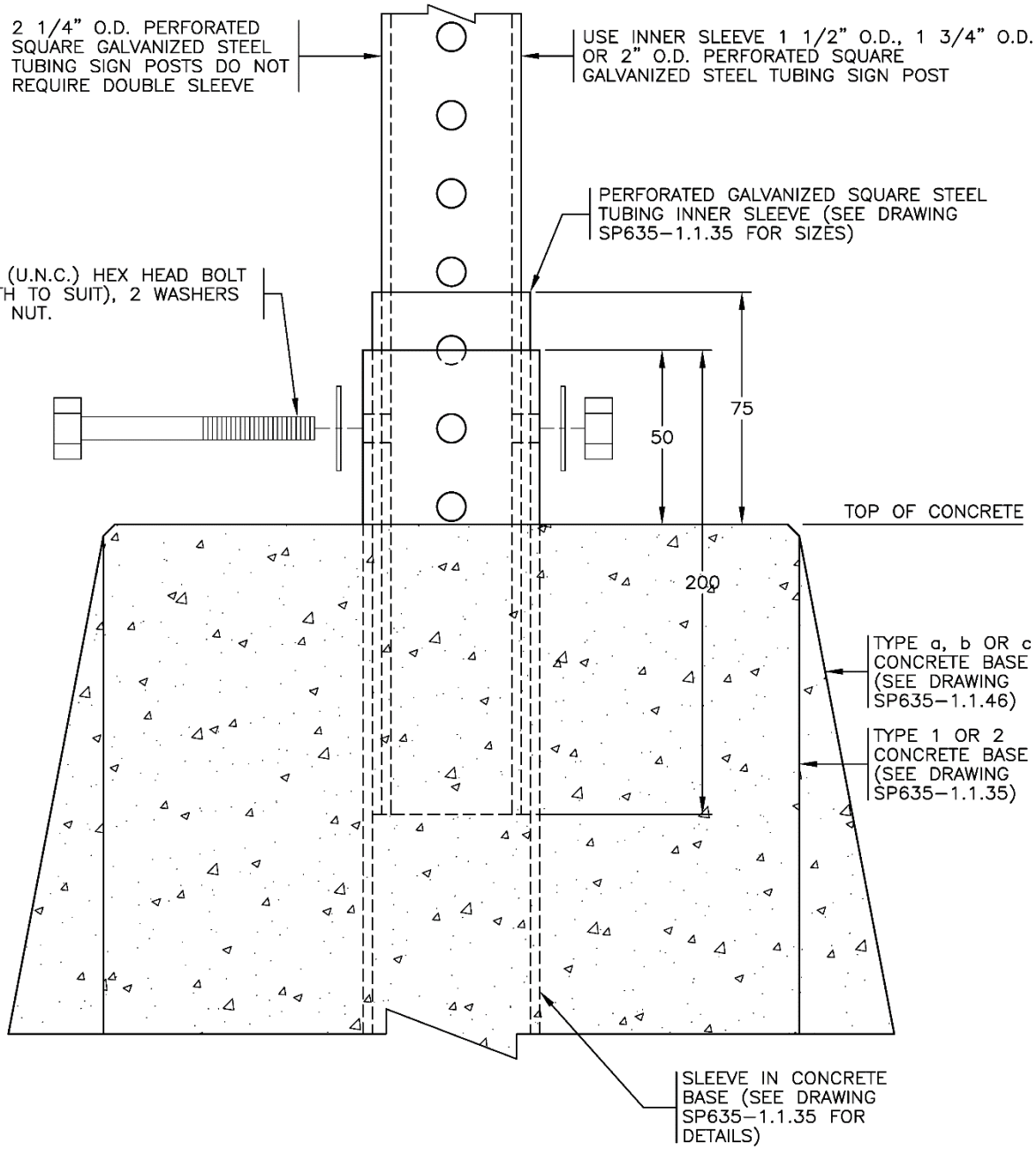
Ministry of Transportation and Infrastructure

No.	Revision	Date
F	GENERAL REVISIONS	APR 20
E	GENERAL REVISIONS	APR 17
D	TRAPEZOIDAL BASE ADDED	APR 05
C	MOUNTING HEIGHTS & OFFSETS REFERED THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS	NOV 98
B	BASE DWG. & LOADING TABLE REFERENCE NOTES REV'D	AUG 96

PERFORATED SQUARE GALVANIZED STEEL SIGN POST INSTALLATION DETAIL (TWO POST)

Date 18/11/94 Approved E.L. (Signature on File) Chief Highway Engineer

SPECIFICATION DRAWING No. SP635–3.6.2



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

ALL NUTS, BOLTS AND WASHERS SHALL BE STAINLESS STEEL

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F	GENERAL REVISIONS	APR 20
E	GENERAL REVISIONS	APR 17
D	REVISION TO SLEEVE HEIGHT	DEC 06
C	TRAPEZOIDAL BASE ADDED	APR 05
B	BASE DRAWING REFERENCES REVISED	AUG 96
A	GENERAL REVISIONS	AUG 95

PERFORATED SQUARE GALVANIZED STEEL SIGN POST INSTALLATION DETAIL

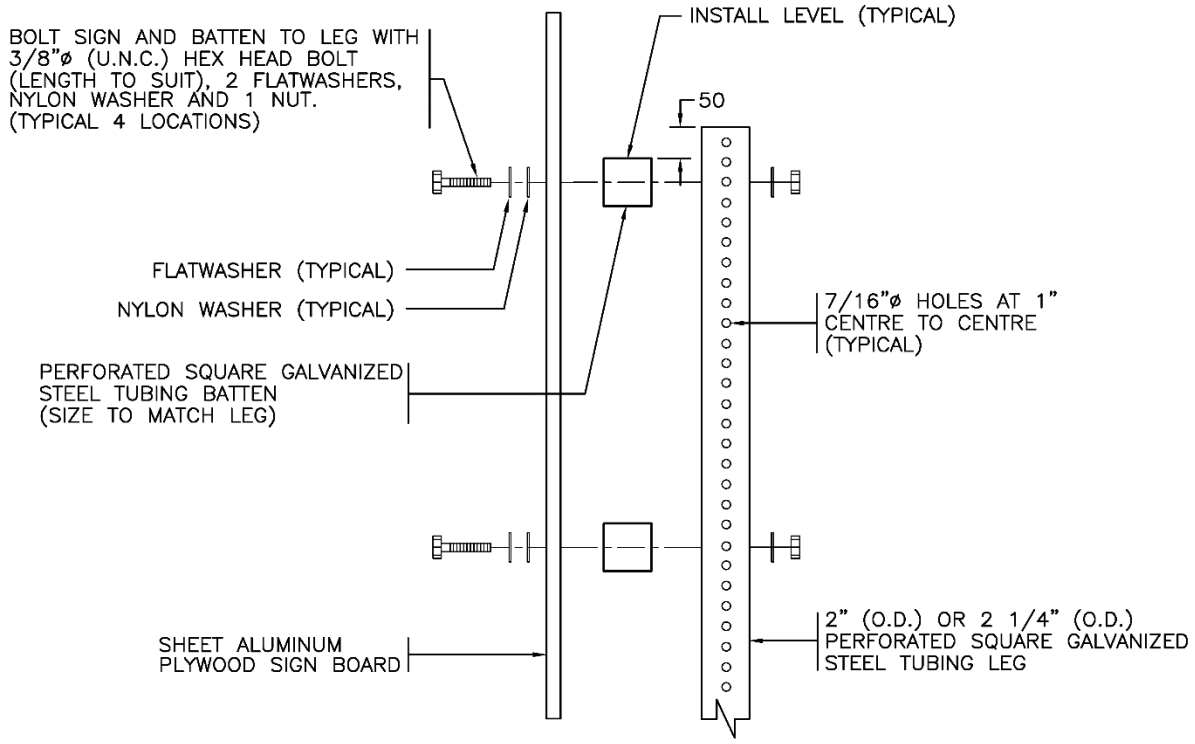
Date Approved

18/11/94 E.L. (Signature on File)

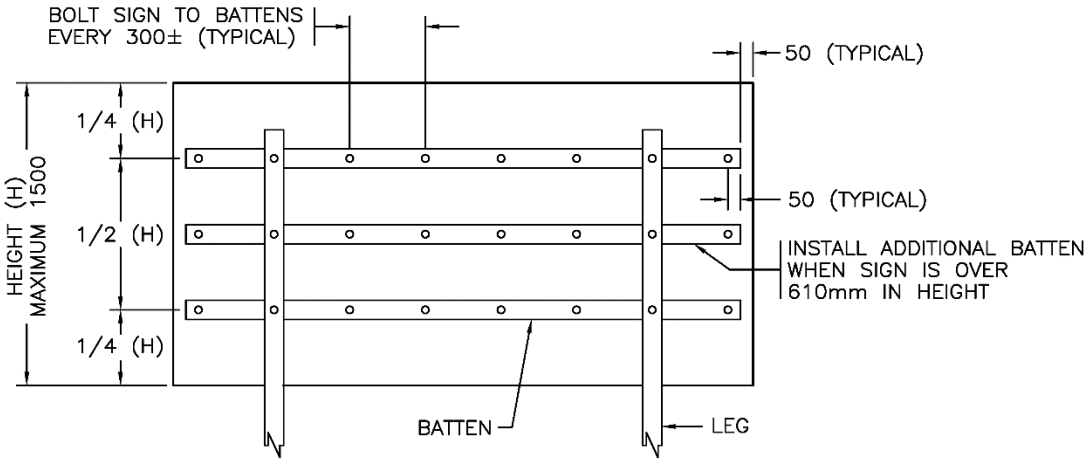
Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-3.6.3



SIDE VIEW



ELEVATION

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

ALL NUTS, BOLTS AND WASHERS SHALL BE STAINLESS STEEL

NOT TO SCALE

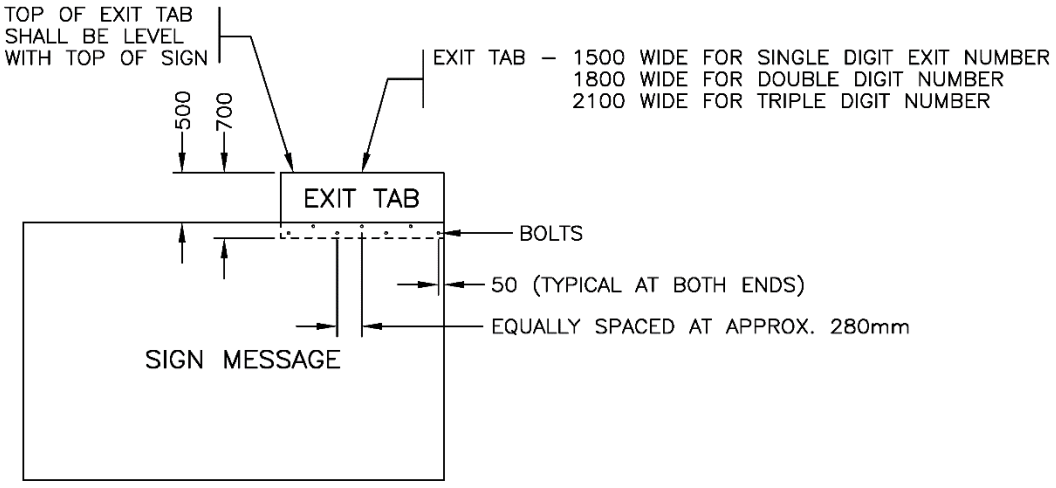
Ministry of Transportation and Infrastructure

No.	Revision	Date
E		
D		
C	GENERAL REVISIONS	APR 20
B	GENERAL REVISIONS	APR 17
A	3rd BATTEN ADDED FOR SIGNS OVER 610mm IN HEIGHT	APR 05

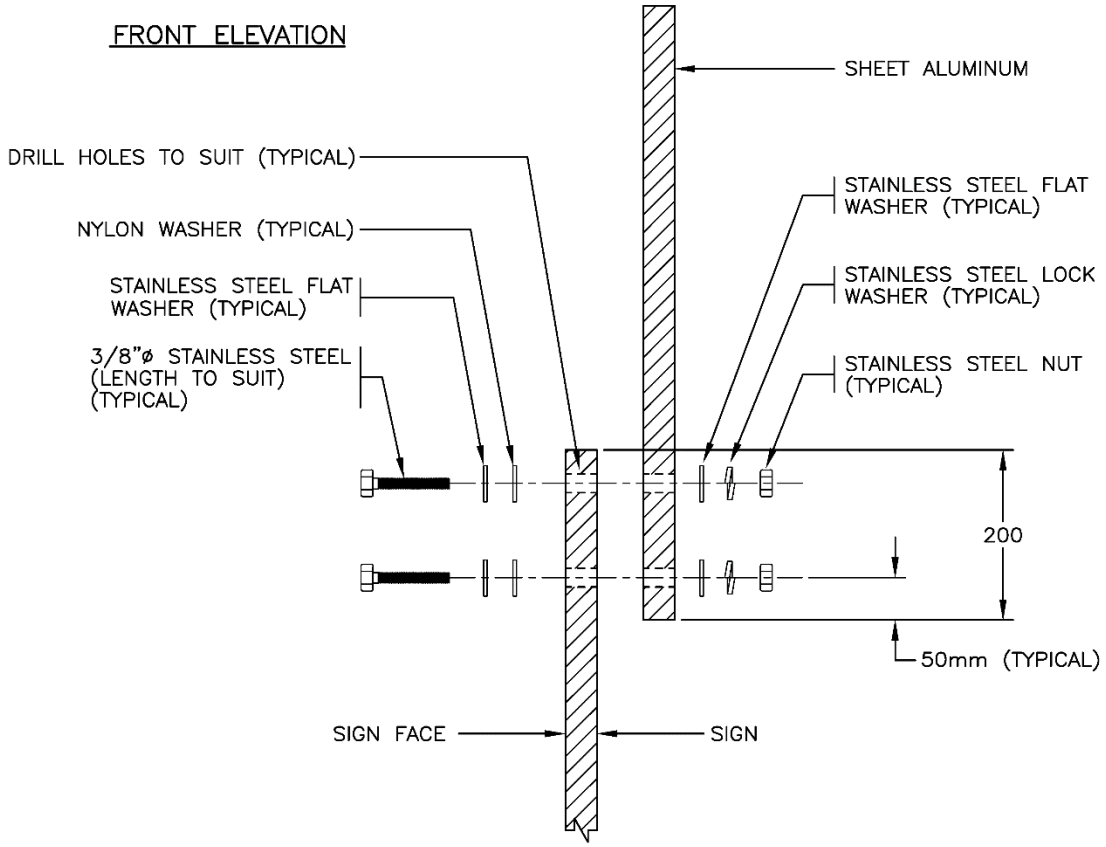
SIGN MOUNTING DETAILS ON PERFORATED SQUARE GALVANIZED STEEL TWO POST SIGN STRUCTURE

Date: **18/11/94** Approved: **E.L.** **(Signature on File)**
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635–3.6.4



FRONT ELEVATION



SIDE VIEW

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	REMOVED PLYWOOD TAB DETAILS	APR 20
C	GENERAL REVISIONS	APR 17
B	GENERAL REVISIONS	APR 05
A	GENERAL REVISIONS	AUG 94

EXIT TAB MOUNTING DETAILS

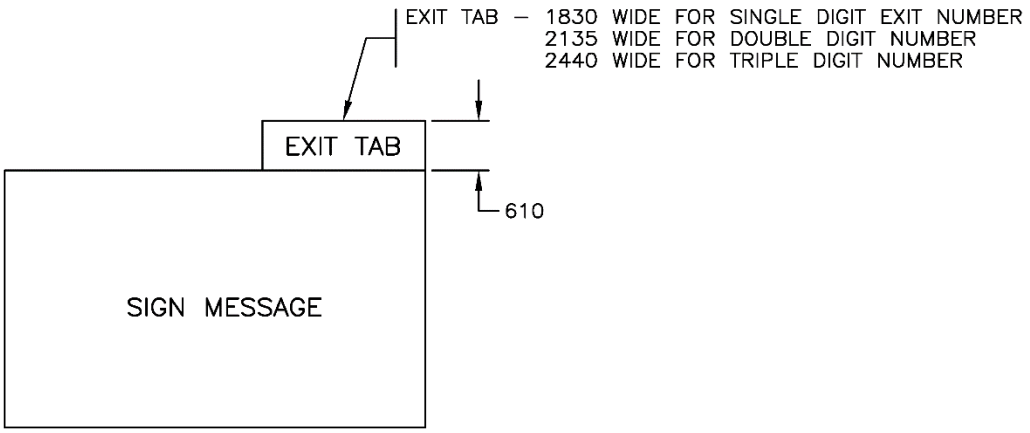
Date Approved

30/09/93 E.L. (Signature on File)

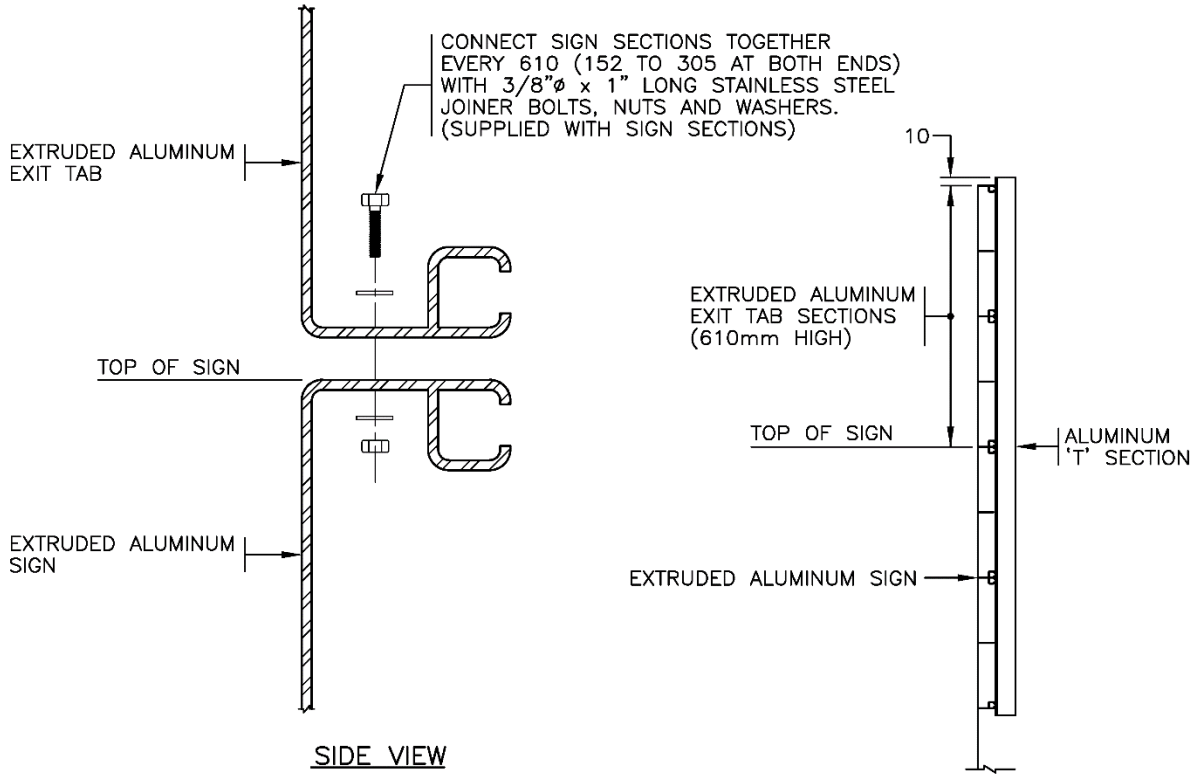
Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–3.7.1



FRONT ELEVATION



SIDE VIEW

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B	GENERAL REVISIONS	APR 17
A	TAB WIDTHS & END HOLE SPACINGS REVISED	AUG 95

EXTRUDED ALUMINUM EXIT TAB MOUNTING DETAILS

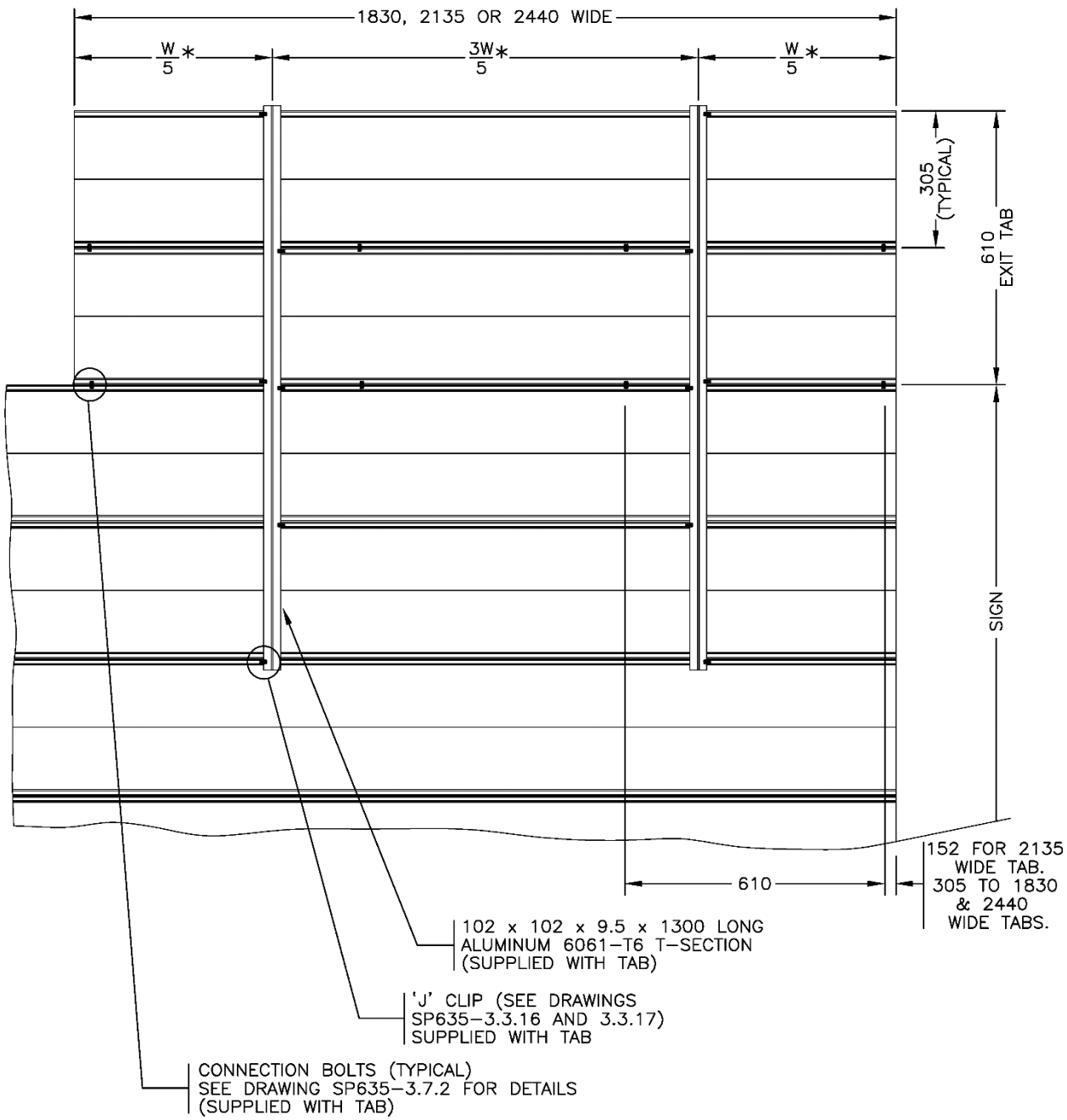
Date Approved

18/11/94 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–3.7.2



REAR VIEW OF SIGN

SEE DRAWING SP635–3.7.2 FOR NOTES AND ADDITIONAL DETAILS.

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	GENERAL REVISIONS	APR 17
B	DRAWING REFERENCE NUMBERS REVISED	AUG 96
A	SIGN WIDTHS & END HOLE SPACING REVISED	AUG 95

EXTRUDED ALUMINUM EXIT TAB MOUNTING DETAILS

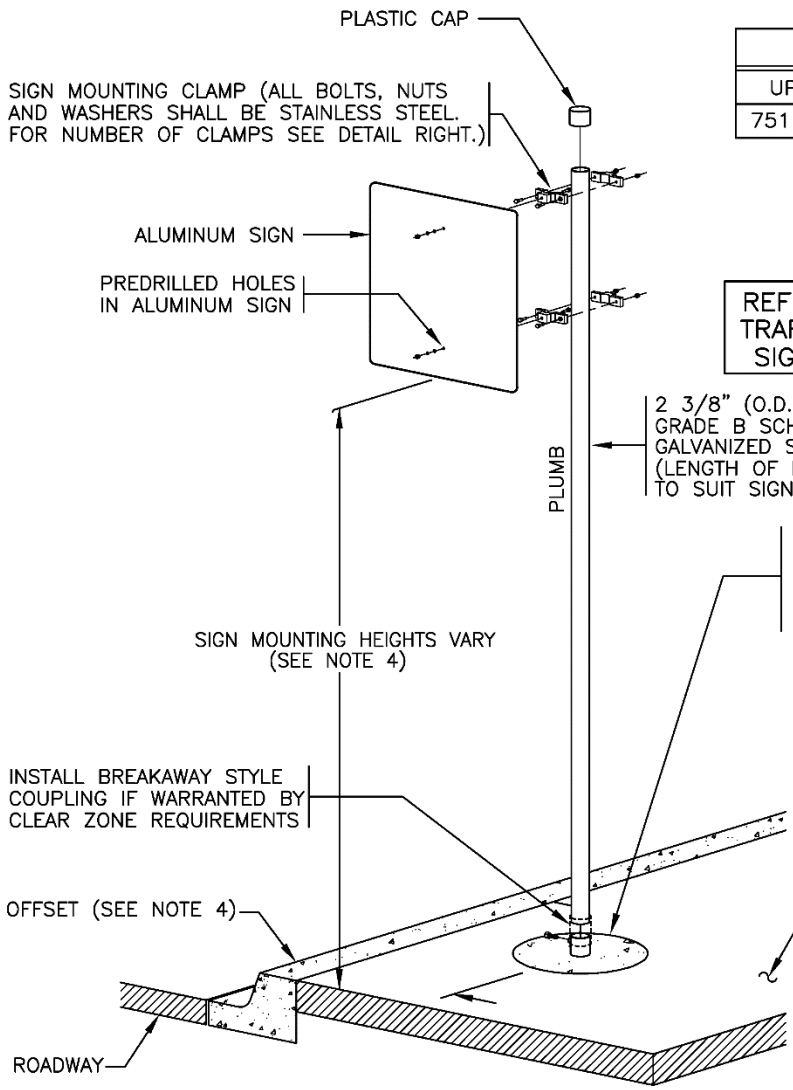
Date Approved

18/11/94 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

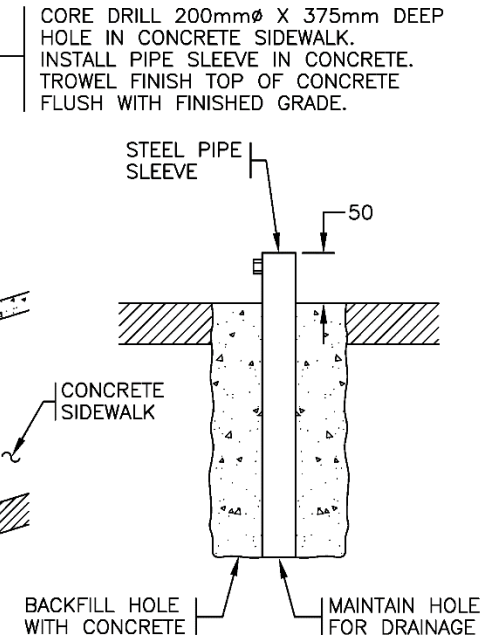
SP635–3.7.3



SIGN HEIGHT	NUMBER OF CLAMPS
UP TO 750 HIGH	2
751 TO 1260 HIGH	3

MAXIMUM SIGN WIDTH
915

REFER TO MINISTRY ELECTRICAL & TRAFFIC ENGINEERING MANUAL FOR SIGN STRUCTURE LOADING TABLE



SLEEVE INSTALLATION DETAIL

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. CONCRETE SHALL HAVE A MINIMUM STRENGTH OF 30MPa PRIOR TO POST INSTALLATION.
4. MOUNTING HEIGHTS & OFFSETS SHALL FALL WITHIN THE RANGES NOTED IN THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS & PAVEMENT MARKINGS.

WHERE INSTALLED IN GRAVEL SHOULDER OR ASPHALT ISLANDS, INSTALL POST ON CONCRETE BASE AS PER DRAWING SP635-1.1.36

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F	GENERAL REVISIONS	APR 20
E	GENERAL REVISIONS	APR 17
D	MAXIMUM SIGN WIDTH REVISED BREAKAWAY COUPLING NOTE ADDED	APR 05
C	MOUNTING HEIGHTS & OFFSETS REFERED THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS	NOV 98

ROUND STEEL SIGN POST INSTALLATION DETAILS

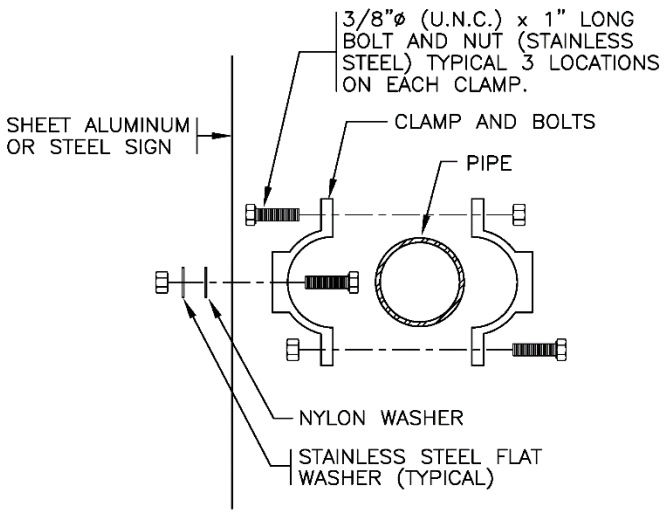
Date Approved

30/09/93 E.L. (Signature on File)

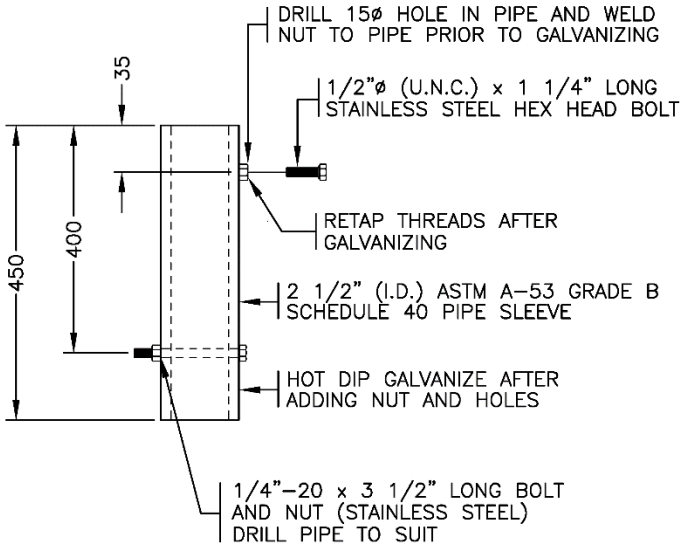
Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-3.8.1



SIGN MOUNTING DETAILS



GALVANIZED PIPE SLEEVE DETAIL

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

NOT TO SCALE

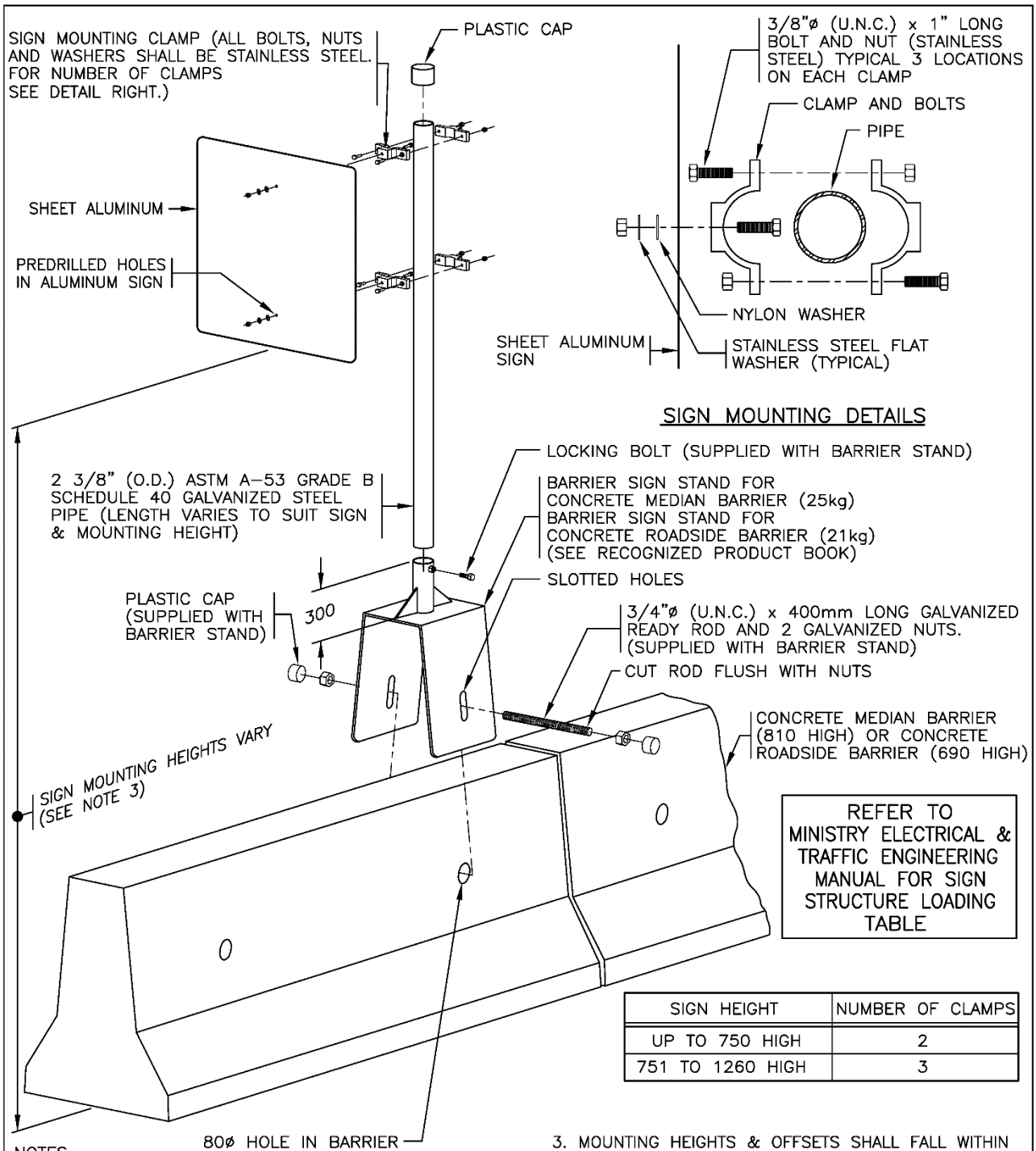
BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B	GENERAL REVISIONS	APR 20
A	GENERAL REVISIONS	APR 17

ROUND STEEL SIGN POST INSTALLATION DETAILS

Date Approved: **18/11/94** **E.L.** **(Signature on File)** Chief Highway Engineer

SPECIFICATION DRAWING No. SP635-3.8.2



REFER TO MINISTRY ELECTRICAL & TRAFFIC ENGINEERING MANUAL FOR SIGN STRUCTURE LOADING TABLE

SIGN HEIGHT	NUMBER OF CLAMPS
UP TO 750 HIGH	2
751 TO 1260 HIGH	3

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. MOUNTING HEIGHTS & OFFSETS SHALL FALL WITHIN THE RANGES NOTED IN THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS & PAVEMENT MARKINGS.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
G	GENERAL REVISIONS	APR 20
F	GENERAL REVISIONS	APR 17
E	REFER TO RECOGNIZED PRODUCT BOOK	APR 05
D	MOUNTING HEIGHTS & OFFSETS REFERED THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS	NOV 98
C	DWG. No. & MOUNTING HEIGHTS REV'D	DEC 96

ROUND STEEL SIGN POST ON BARRIER SIGN STAND INSTALLATION DETAILS

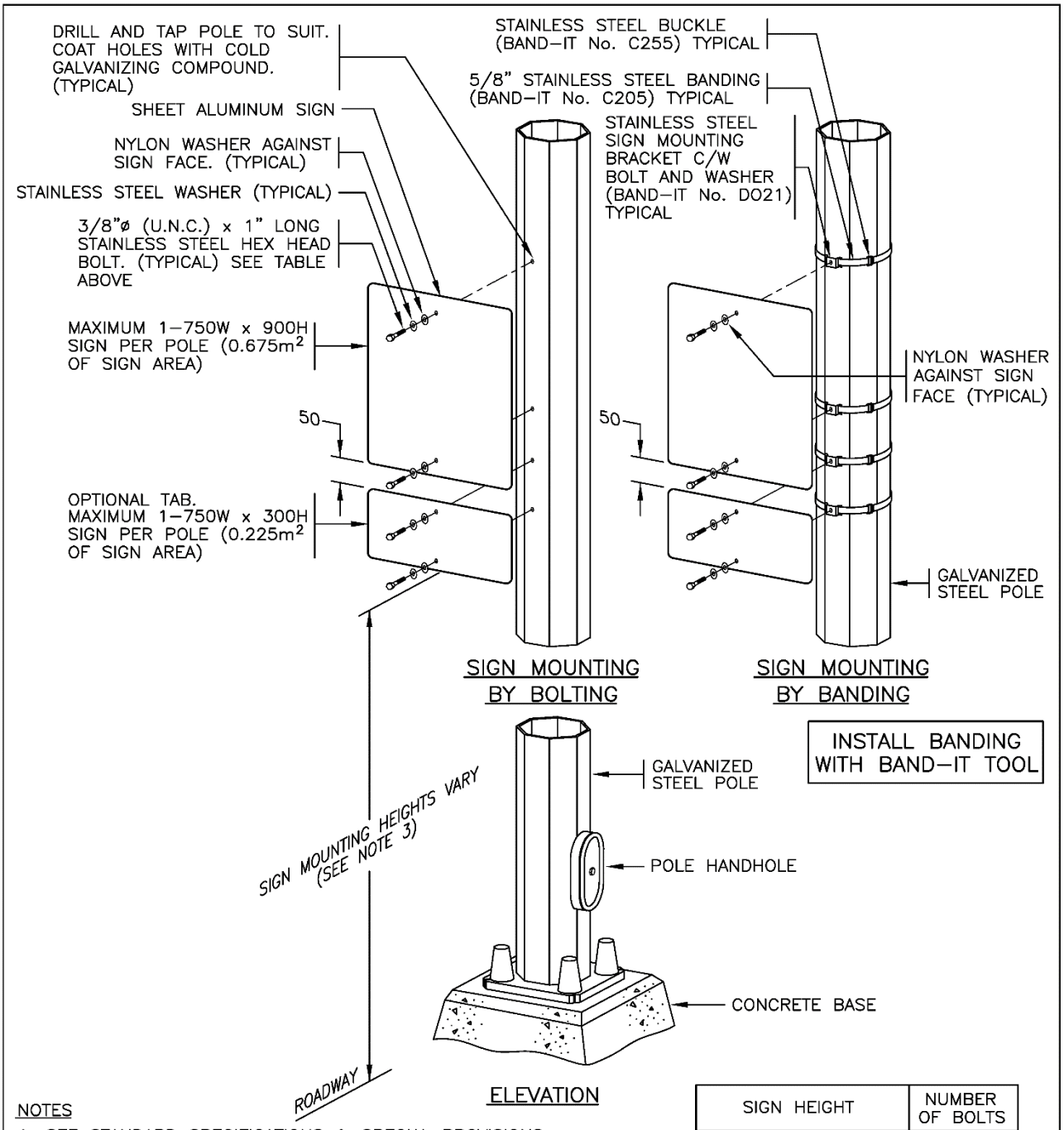
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-3.8.3



SIGN HEIGHT	NUMBER OF BOLTS
UP TO 750 HIGH	2
751 TO 1200 HIGH	3

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. MOUNTING HEIGHTS & OFFSETS SHALL FALL WITHIN THE RANGES NOTED IN THE MINISTRY MANUAL OF STANDARD TRAFFIC SIGNS & PAVEMENT MARKINGS.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
I	GENERAL REVISIONS	APR 20
H	GENERAL REVISIONS	APR 17
G	MINISTRY APPROVAL NOTE REVISED	APR 05
F	MINISTRY APPROVAL NOTE ADDED	NOV 01
E	BANDING SIGN MOUNTING ADDED	SEPT 00

SIGN INSTALLATION ON THE SIDE OF POLES

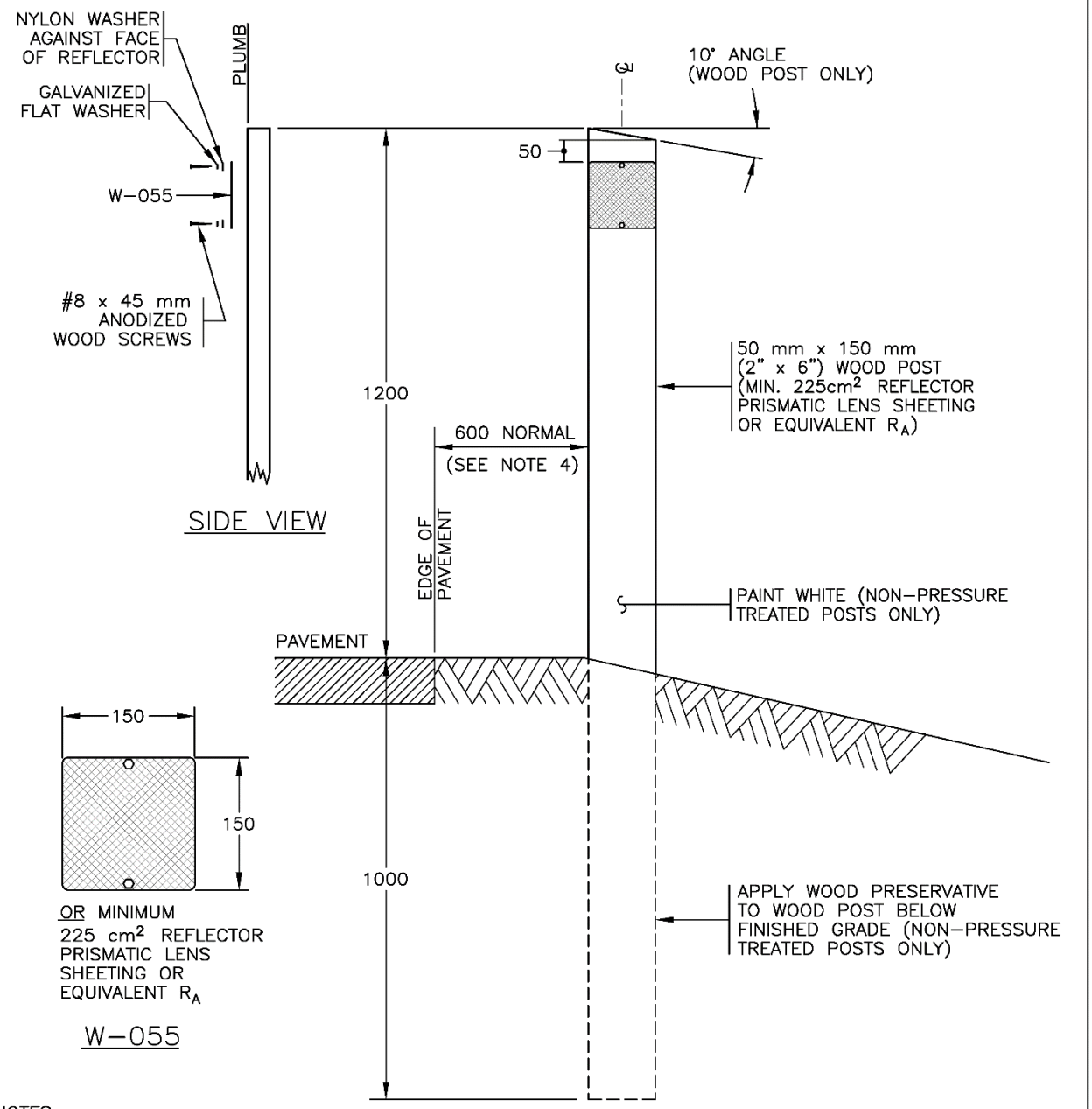
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–3.9.1



NOTES

1. SEE THE MINISTRY'S "MANUAL OF STANDARD TRAFFIC SIGNS AND PAVEMENT MARKINGS" TABLE 7.4a-b FOR DELINEATOR SPACING ON HORIZONTAL CURVES.
2. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
3. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
4. POST PLACED A MINIMUM OF 1.0 m TO A MAXIMUM OF 3.6 m FROM EDGE OF TRAVELLED LANE.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
G	NOTE 1 UPDATED	APR 20
F	GENERAL REVISIONS	APR 17
E	PRESSURE TREATED POSTS ADDED	APR 05
D	SIZE & NUMBER OF REFLECTORS	NOV 01
C	SIGN NUMBER REVISED	AUG 96
B	OFFSETS ADDED	AUG 95

WOOD DELINEATOR POST INSTALLATION DETAILS

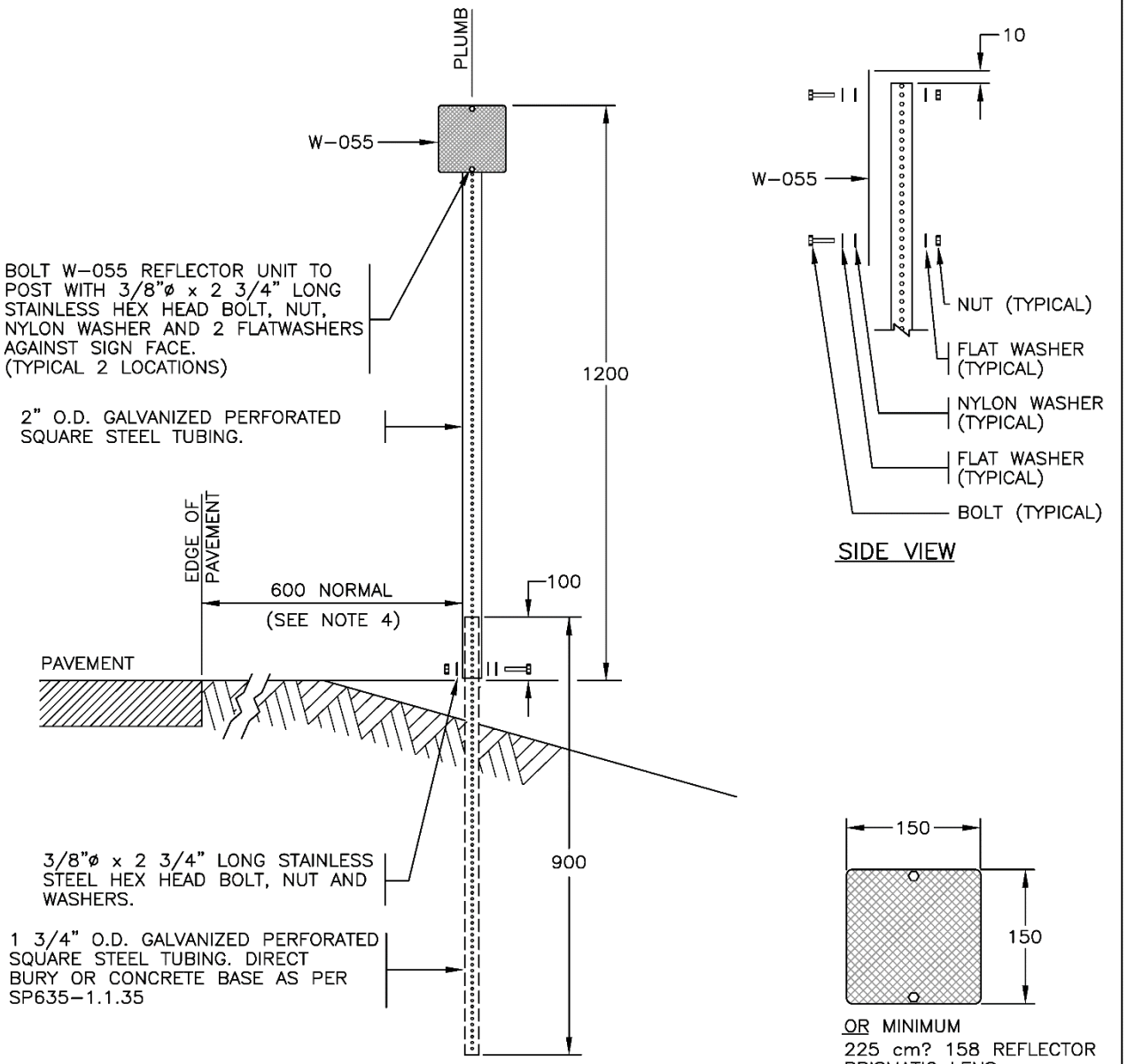
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-3.10.1



W–055

NOTES

1. SEE THE MINISTRY'S "MANUAL OF STANDARD TRAFFIC SIGNS AND PAVEMENT MARKINGS" TABLE 7.4a–b FOR DELINEATOR SPACING ON HORIZONTAL CURVES.
2. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
3. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
4. POST PLACED A MINIMUM OF 1.0 m TO A MAXIMUM OF 3.6 m FROM EDGE OF TRAVELLED LANE.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
G	NOTE 1 UPDATED	APR 20
F	GENERAL REVISIONS	APR 17
E	GENERAL REVISION	APR 05
D	SIZE & NUMBER OF REFLECTORS	NOV 01
C	OFFSET & POST LENGTHS REVISED	OCT 99
B	W–055 SIGN REFERENCES REVISED	AUG 96

PERFORATED SQUARE STEEL TUBING DELINEATOR POST INSTALLATION DETAILS

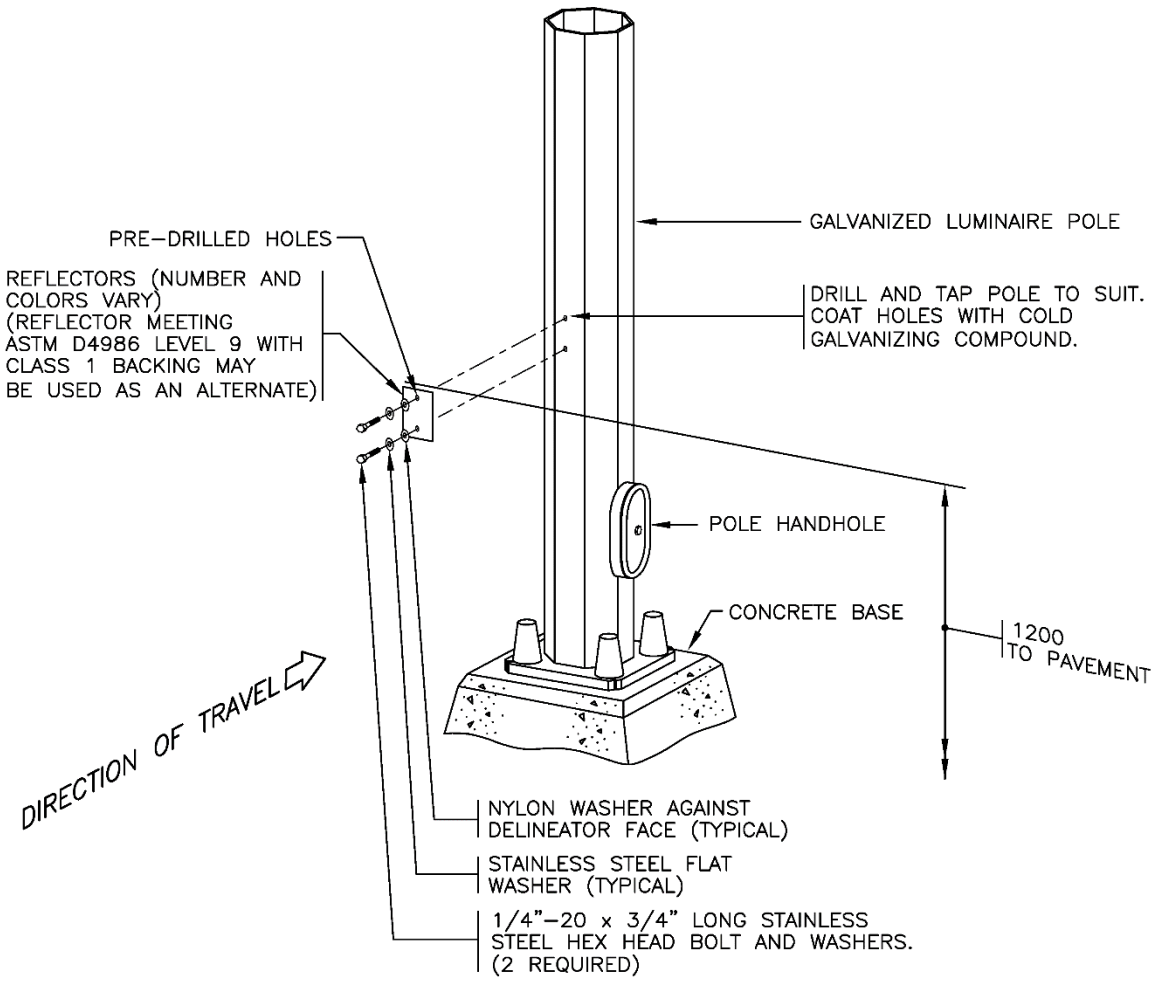
Date Approved

18/11/94 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–3.10.2



ASSEMBLY DETAIL

NOTES

1. NUMBER, COLOUR, SIZE AND SHAPE OF REFLECTORS VARIES. SEE THE MINISTRY'S MANUAL OF STANDARD TRAFFIC SIGNS AND PAVEMENT MARKINGS FOR MORE INFORMATION.
2. DELINEATOR MANUFACTURER MUST MEET MINISTRY APPROVAL.
3. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
5. INSTALL REFLECTOR ON LUMINAIRE POLE (OR SIMILAR) IF DELINEATOR TO BE POSITIONED WITHIN 5m OF POLE (OR SIMILAR).

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

NOT TO SCALE

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 20
C	GENERAL REVISIONS	APR 17
B	NOTE 5 ADDED	APR 05
A	GENERAL REVISION	AUG 94

PLASTIC DELINEATOR INSTALLATION ON POLE

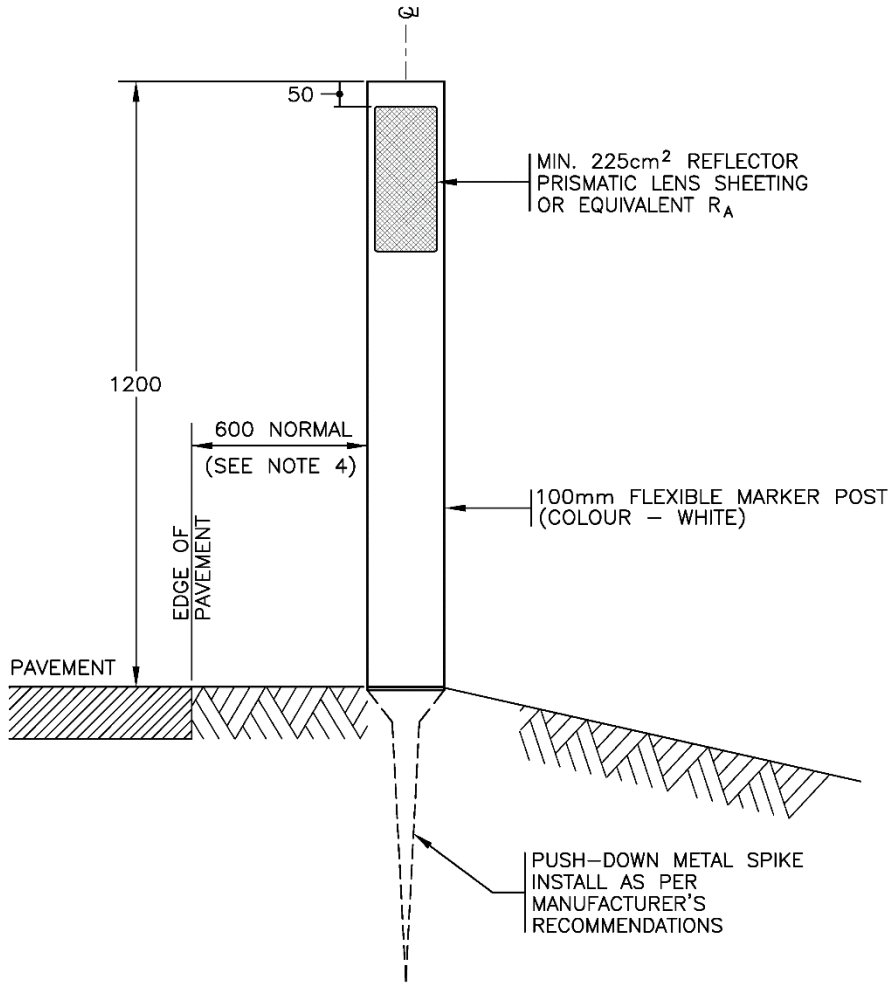
Date Approved

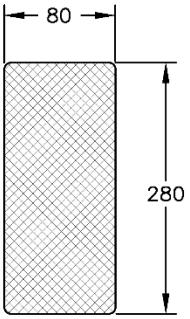
30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–3.10.3





OR MINIMUM
225 cm² REFLECTOR
PRISMATIC LENS
SHEETING OR
EQUIVALENT R_A

W–055

NOTES

1. SEE THE MINISTRY'S "STANDARD TRAFFIC SIGNS AND PAVEMENT MARKING MANUAL" TABLE 7.4a–b FOR DELINEATOR SPACING ON HORIZONTAL CURVES.
2. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
3. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
4. POST PLACED A MINIMUM OF 1.0 m TO A MAXIMUM OF 3.6 m FROM EDGE OF TRAVELLED LANE.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B	NOTE 1 UPDATED	APR 20
A	GENERAL REVISIONS	APR 17

FLEXIBLE DELINEATOR POST INSTALLATION DETAILS

Date Approved: **20/04/05 D.N. (Signature on File)**
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635–3.10.4

SECTION 700
WILDLIFE EXCLUSION FENCING

DESCRIPTION

700.01 Scope – This Section covers the construction of wildlife exclusion fencing with galvanized wire mesh, pressure treated wood posts and/or galvanized steel posts, one-way gates, lockable human access gates and double swing gates and, with reference to SS Drawings of the SP700 series, is intended to specify acceptable standards and some optional features as may be required by the Special Provisions.

Alternative construction may be called for by the Drawings, the Special Provisions or instructions of the Ministry Representative, and alternative methods may be acceptable upon submission to the Ministry Representative.

MATERIALS

700.10 Post Type – Posts may be wood or steel, except that steel posts shall be used:

(a) where it is not possible to attain the required embedment for wood posts, such as where bedrock or boulders obstruct the post hole;

(b) on ground slopes steeper than 1.5 horizontal to 1 vertical (34°);

(c) in wetlands, in areas frequently under water, or in areas of soft organic soils; and

(d) where it is determined that wood posts are unsuitable for the existing ground conditions, topography or other environmental factors.

700.11 Treated Wood Materials

700.11.01 General – Wood posts, braces and other timbers shall be supplied by the Contractor in accordance with the requirements of SS 909.

All wood products used in wildlife exclusion fence construction shall be preservative treated and meet the requirements of SS 908.

No cutting of preservative treated wood posts will be permitted without authorization of the Ministry Representative.

When cutting is authorized, the cut must be only at the top of the posts.

All cuts, holes, and superficial damage shall be field treated immediately with a preservative in accordance with SS 908.

700.11.02 Posts

(a) Round Wood Post Material – Treated Round Wood posts shall be prepared from straight peeled Jack Pine or Lodgepole Pine.

(b) Dimensional Tolerances – In addition to the requirements in SS 909, treated round wood posts shall have the following dimensions, each within a 2% tolerance:

(i) Length shall be 4300 mm or as necessary to comply with the applicable SS 700 SP Drawings, for the fence type specified and the soil type encountered;

(ii) minimum tip diameter of 150 mm; and

(iii) maximum butt diameter of 200 mm.

(c) Bottom Tapers – The bottoms of the round wood posts must be tapered to a rounded end for driveability.

700.12 Not Used

700.13 Steel Posts and Steel Components – All steel posts and steel components shall be hot-dipped galvanized in accordance with ASTM A153 or ASTM A123, as applicable.

Steel posts shall generally be Schedule 40, except shall be Schedule 80 where located in wetlands, in areas frequently under water, or in areas of soft organic soils.

The steel posts will not have an outside diameter less than 73 mm. The length of steel posts may vary between 3560 mm and 4500 mm according to installation conditions as shown on the Drawings.

Upon approval by the Ministry Representative, short steel posts may be lengthened, by a maximum of 1000 mm, by a welded extension. All welds shall be inspected by the Ministry Representative. Welds will be ground as necessary to achieve a smooth, bare metal surface and immediately coated per SS 700.14.

700.14 Touch-up Treatment for Galvanized Metal Surfaces – Touch-up treatment for field-damaged galvanizing of steel posts and braces shall be two coats of an organic, zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers and applied on a thoroughly cleaned surface.

700.15 Fabric and Wire – The fence fabric will be 2.44 m (8 ft) high with 150 mm (6 in) horizontal and graduated vertical spacing, 2.5 mm (12.5 gauge) high tensile wire with a twisted friction type joint at each horizontal/vertical contact point. The fence fabric will have a minimum of 20 horizontal wires, with graduated vertical spacing ranging

from 7.62 cm (3 inches) at the bottom to 17.78 cm (7 inches) at the top. The fence fabric and wire will be galvanized in accordance with ASTM A123, to a retention of not less than 240 g/m^2. The horizontal wires will have a minimum tensile strength of 1234 MPa (179 ksi).

All wire used for braces and other use shall be galvanized to minimum Class 3 in accordance with ASTM A641.

700.15.01 Fence Fabric in Confined Locations – Where space limitations do not permit the handling of the full 2.44 m (8 ft) section of fence, it will be permissible, where approved by the Ministry Representative, to use two, 1.22 m (4 ft) sections of fencing. These sections will be connected by galvanized steel compression hog rings every 150 mm or joined longitudinally with a continuous 3 mm (11 gauge) galvanized wire woven through the mesh and wrapped every 150 mm.

700.16 One-Way Gates – Gates shown on the Drawings shall be of the prefabricated type indicated on SS Drawings SP700-12.1 to SP700-12.6 and as specified in SS 316.11 and/or in the Special Provisions. The one-way gates must be test assembled in shop, adjusted and lubricated to swing freely without binding, and quietly shut from any position. All metal components of gates shall be galvanized.

700.16.01 One-Way Gate Tines – Tines shall be fabricated from AISI 4140 high tensile steel to avoid animals or people bending them. Subsequent to cold-bending the tines, and prior to hot-dipped galvanizing, the material shall be heat treated and stress relieved to achieve a Rockwell C Hardness (HRC) value of 30-35. The vertical gap between sections of tines must not exceed 200 mm. The horizontal gap between the left and right sections of tines of the one-way gates must not exceed 100 mm.

All metal in one-way gates must be galvanized. Acid bath treatment of the tines prior to galvanization shall be controlled to minimize potential weakening of welds.

700.17 Lockable Human Access Gates – Gates shown on the Drawings shall be of the prefabricated type indicated on SS Drawings SP700-14.1 and SP700-14.2. The lockable human access gates must be adjusted and lubricated to swing and shut freely without binding in the field. Lock assemblies must be adjusted to work in the field. All metal components of gates shall be galvanized.

700.18 Double-Swing Gates – Gates shown on the Drawings shall be of the prefabricated double-swing type indicated on SS Drawings SP700-13.1 and SP700-13.2. The human access gates must be adjusted and lubricated to swing and shut freely without binding in the field. Lock assemblies must be adjusted to work in the field. All metal components of gates shall be galvanized.

CONSTRUCTION

700.20 Provision of Fencing – Fencing shall be carried out at the locations and as shown on the Drawings with the materials to the height, spacing and with accessories all in accordance with the details indicated on the Drawings, Standard Specifications, and Special Provisions or to the direction of the Ministry Representative.

All material shall be supplied by the Contractor, except where supply in whole or in part by the Ministry, DDP (Incoterms© 2020: Delivery Duty Paid). the Contractor's job site yard or Ministry's yard, is specified in the Special Provisions.

Construction shall be carried out with all labour, tools, equipment and incidentals supplied by the Contractor, as necessary, to complete all fencing work in accordance with good work practice.

700.21 Clearing & Grading – Prior to commencing fencing work, both sides of the fence line must be free of all clearing and grubbing debris. All trees, other than those required by the Ministry Representative to remain, and all brush and other obstacles which interfere with the construction and maintenance of fencing and not removed by the normal clearing operations, must be removed.

The cleared and graded area must be a minimum of 3 m wide on each side of the fence to permit access for fence repairs and maintenance, unless a lesser width is permitted by the Ministry Representative. The graded area shall have a 2-5% cross-fall towards the open side of the slope to provide drainage. No fill shall be placed in any drainage path that crosses the fence line.

The ground line for the fence should be smooth and continuous for a minimum of 1 m on both sides of the fence. Minor ground undulations shall be corrected to obtain a smooth uniform grade, but appreciable grade depressions may be backfilled only with the permission of the Ministry Representative.

The site shall be left in a smooth and tidy condition.

700.22 Setting Out and Connections to Existing Fences – Fence line, as shown on the Drawings, generally will be located a minimum of 3 m from the right-of-way boundary, unless a lesser distance is permitted by the Ministry Representative.

In areas of heavy snow, where the 10 year average total annual snowfall is greater than 2 m, as identified in Special Provisions, the fence line must be offset a minimum of 4 m from the edge of the highway shoulder to reduce damage from snowplows. Otherwise, the fence line must be offset a minimum of 3 m from the edge of the highway shoulder.

Post installation in fill material or minimum overburden shall be carried out according to the Ministry Representative's directions. Where it is not possible to drive or set wood posts to proper depth or to relocate same along

the fence line, steel fence posts as specified or, where permitted, multiple wood post and brace assemblies shall be substituted.

Existing fences shall be connected to new fences with posts and braces for tensioning fencing wire in every direction in accordance with the SP700 series of SS Drawings.

700.23 Post Spacing – Except in areas of heavy snow, line posts shall be spaced 5 m apart measured horizontally. It will be permissible to move a post, up to 0.3 m ahead or back along the fence line, to avoid an obstruction preventing advancement of the post hole, provided that the average spacing does not exceed 5 m.

In areas of heavy snow, where the 10-year average total annual snowfall is greater than 2 m, as identified in Special Provisions, the line post spacing must be reduced to 3.5 m to reduce fence fabric sagging and tearing.

700.23.01 Installation of Wood Posts – Wood posts shall be installed plumb and to a depth of 1500 mm, as indicated on the Drawings. Posts may be either driven or set in excavated holes, provided that a rigid installation is achieved, capable of withstanding a horizontal load of 32 kg (70 lb), applied 1.5 m above the ground, in any direction in the horizontal plane, without any movement in excess of 25 mm. When a wood post is set in an excavated hole, the soil around the wood post must be compacted to the satisfaction of the Ministry Representative. The horizontal load test must be conducted after the post is installed but before the fabric wire is attached. It is the responsibility of the Contractor to conduct and record the horizontal load test. The results of the horizontal load test must be submitted to the Ministry Representative.

Where the slope of the terrain along the fence line approaches 1.5H:1V (34°), embedment of wood posts may be reduced to a minimum of 1200 mm, with permission of the Ministry Representative. Where fence gradients are steeper than 1.5H:1V (34°), steel posts must be used, as indicated on the Drawings.

700.23.02 Concrete – All concrete used as post anchorage backfill shall be well-consolidated Class C concrete conforming to SS 218 Table 218-A, or a non-shrink grout, minimum compressive strength of 20 MPa at 28 days, approved by the Ministry Representative.

700.23.03 Installation of Steel Posts – Steel line posts, end posts, corner posts and brace posts embedded in solid rock shall be set in concrete or non-shrink grout.

Where bedrock is encountered on steep slopes (steeper than 1.5H:1V; 34°) corrosion resistant concrete anchors that have manufacturer's specifications that recommend use for installations in accordance with the available concrete edge distance and that are acceptable to the Ministry Representative. Where wood posts are used, wood posts shall be located as close to the abutments as possible.

700.24 Fence Ties – Fencing shall be tied into structures, gates and existing fencing as staked in the field or as directed by the Ministry Representative. Fencing ties will at no time leave a gap greater than 100 mm.

Care must be taken to ensure that the fencing ties and post installation does not compromise the effectiveness of the adjacent structures.

At ungulate guards and bridges, posts may be wood or steel as permitted by the Ministry representative. Where steel posts are used, steel posts shall be bolted to the concrete abutments using 20 mm diameter galvanized or stainless stell corrosion-resistant concrete anchors that have manufacturer's specifications that recommend use for installations in accordance with the available concrete edge distance and that are acceptable to the Ministry Representative. Where wood posts are used, wood posts shall be located as close to the abutments as possible.

700.25 Brace Panels – Brace panels will include intermediate brace panels, double intermediate brace panels consisting of two intermediate brace panels back to back and end post panels.

Brace panels shall be constructed and installed as shown on the SS Drawings SP700-01 and SP700-02. Cross wires shall be twisted to provide suitable tension, in the manner illustrated on the SS Drawing SP700-01.

Bracing wire must be galvanized and a minimum of 9.75 gauge.

The spacing between adjacent intermediate brace panels, and between intermediate brace panels and end post panels, will not be more than 54 m, unless otherwise permitted by the Ministry Representative.

Intermediate brace panels shall be installed where necessary to meet the foregoing requirement, and also where required by the Ministry representative at changes in vertical or horizontal alignment.

Double brace panels will be used at corners and other sharp changes in the vertical or horizontal alignment, and at any other locations where they are, in the judgement of the Ministry Representative, required to maintain the integrity of the fence.

End post panels will be installed where the fence ties in to structures or gates, at ungulate guards and at any other termination of the fence. The requirement for brace panels at one-way wildlife gates is shown on SS Drawing SP700-12.1.

700.26 Fence Fabric Installation – The fence fabric will be fastened to posts on the surface facing away from the highway right-of-way, except where the fence is located adjacent to concrete barriers and overpasses.

For wood posts, the wire fence fabric shall be stapled to each post, using a maximum vertical spacing of 150 mm including both the top and bottom wires. Staples will be

barbed and galvanized steel, 3.5 mm in diameter with a driven length of at least 50 mm.

The fence fabric will be attached to steel posts with a minimum of four (4) galvanized muffler clamps (e.g. MC12300 P type); one on the top strand, one 0.60 m below the top strand, one 1.20 m below the top strand, and one on the bottom wire of the fence fabric. Intermediate connections will be made every 300 mm along the post with 3.5 mm galvanized wire twisted to form a tight connection.

The wire fence fabric shall be tensioned to provide a uniform pull in order to minimize distortion of the fabric. Each run of fence fabric between brace panels will be tensioned before staples are set or clamps tightened.

The tension of the fence fabric will be considered adequate when the fabric cannot be pulled more than 100 mm out of line with a 13.6 kg (30 lb) pull at any point from top to bottom between the posts. The allowable 100 mm will include any deflection of the post, should this occur.

The fence fabric will be as close to the ground as possible, but in any case, the vertical distance between the bottom strand of wire and the existing ground will nowhere exceed 150 mm. Unerodible, clean fill material, neatly trimmed, will be added to obtain this clearance, where practical.

Where the fence crosses gullies or drainage channels, a specially fabricated section of fence will be cut to fit the opening and will be fastened to the bottom wire and the streambed as shown on the Drawings. Alternatively, but only where required by the Ministry Representative, culvert pipe will be installed at specific crossings in accordance with the Special Provisions.

700.27 Connection Treatments at Structures – Fences must be securely attached to walls, abutments, ungulate guards and other structures to ensure stability of the fences and to prevent animals from passing between the fence and the structure. The fence must be attached as per SP700-11 using concrete anchors as per SS 700.24. The fence must be securely attached to rock faces in accordance with SS Drawing SP700-04.

700.28 Gates – Hardware shall be securely attached to permit the gate to open correctly and prevent the easy removal of the gate and hardware. Hinges shall be installed to permit the gate to swing back one-way against the fence. Locking hardware shall be of the type specified in SS Drawing SP700-13.2.

700.29 Lockable Human Access Gates – Access must be provided to allow inspection access from the highway of all points of all bridges, both ends of culverts greater than 2 m in diameter, retaining walls over 2 m in height, tunnels, and farm and wildlife crossing structures.

Lockable human access gates must be located where they can be safely used, avoiding cliffs, steep slopes, swamps, areas frequently under water, and where unstable ground conditions exist. The gates must be located within 25 m of the structures and culverts they provide access to, unless a greater distance is permitted by the Ministry Representative. The gates shall be of the type specified in SS Drawing SP700-14.1.

Unless otherwise permitted by the Ministry Representative, the minimum number of lockable human gates required at specific features are:

(a) **Culverts greater than 2 m in diameter –** One (1) gate for each fence located adjacent to the culvert.

(b) **Retaining walls greater than 2 m in height**:

(i) **Retaining walls less than 200 m long:** One (1) gate where the wildlife exclusion fencing abuts the wall.

(ii) **Retaining walls greater than 200 m long:** One (1) gate at each location of the wall where the wildlife exclusion fencing abuts the wall.

(c) **Tunnels –** One (1) gate at each tunnel portal.

(d) **Bridges –** One (1) gate for each location where the wildlife exclusion fencing abuts a bridge. The total number of gates required for a bridge may be reduced by the Ministry Representative if sufficient access, during median flow of water conditions, can be provided and maintained with fewer gates.

QUALITY ASSURANCE

700.30 Quality Assurance – The Ministry Representative must be provided a reasonable opportunity to conduct acceptance testing.

The Contractor will disassemble and relocate any elements of the fence necessary for testing at the discretion of the Ministry Representative. If the test of an element is successful, the Ministry shall pay all costs for the replacement of the materials and reconstruction of the element tested.

The Ministry Representative may require any element not tested, or failing the test, to be replaced and retested.

When the Ministry Representative requests a test of an element of the fence and the test fails, all costs will be to the Contractor's account, including the replacement of the materials and reconstruction of the element tested, and all costs associated with the retesting.

700.30.01 Removal and Replacement at Contractor's Expense – The Ministry Representative may require any fencing materials that do not meet the Ministry's specifications to be removed and replaced at the Contractor's expense.

The Ministry Representative may require any constructed fencing or gates that do not meet the Ministry's specifications to be removed and replaced at the Contractor's expense.

MEASUREMENT

700.40 Fencing – Fencing will be measured by the LINEAL METRE. Measurements will be made parallel to the top wire of complete fencing, including any tensioning assemblies, but excluding gate openings.

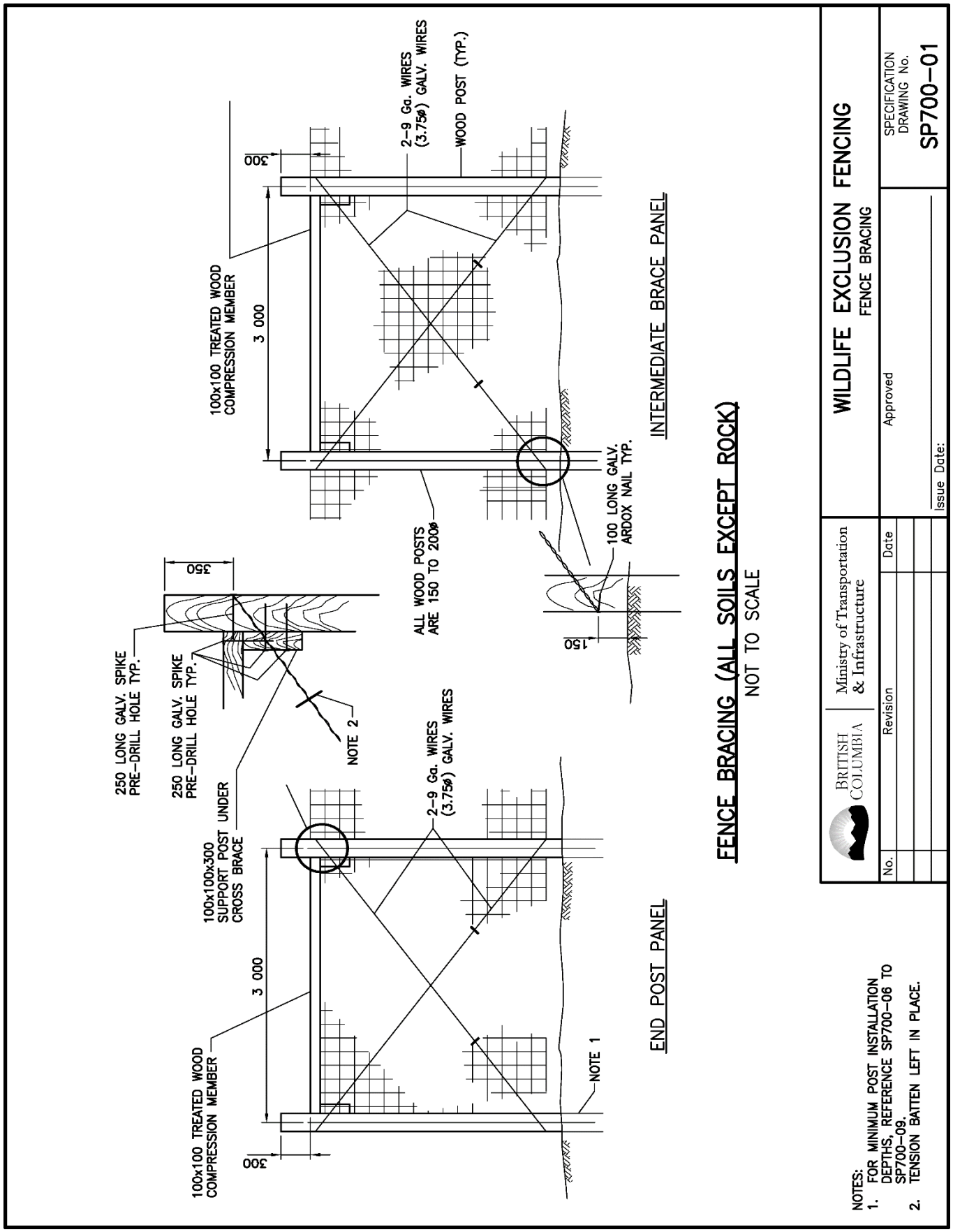
700.41 One-Way Gates, Lockable Human Access Gates and Double Swing Gates – Gates will be measured by the unit for EACH type and size furnished and/or installed complete in place.

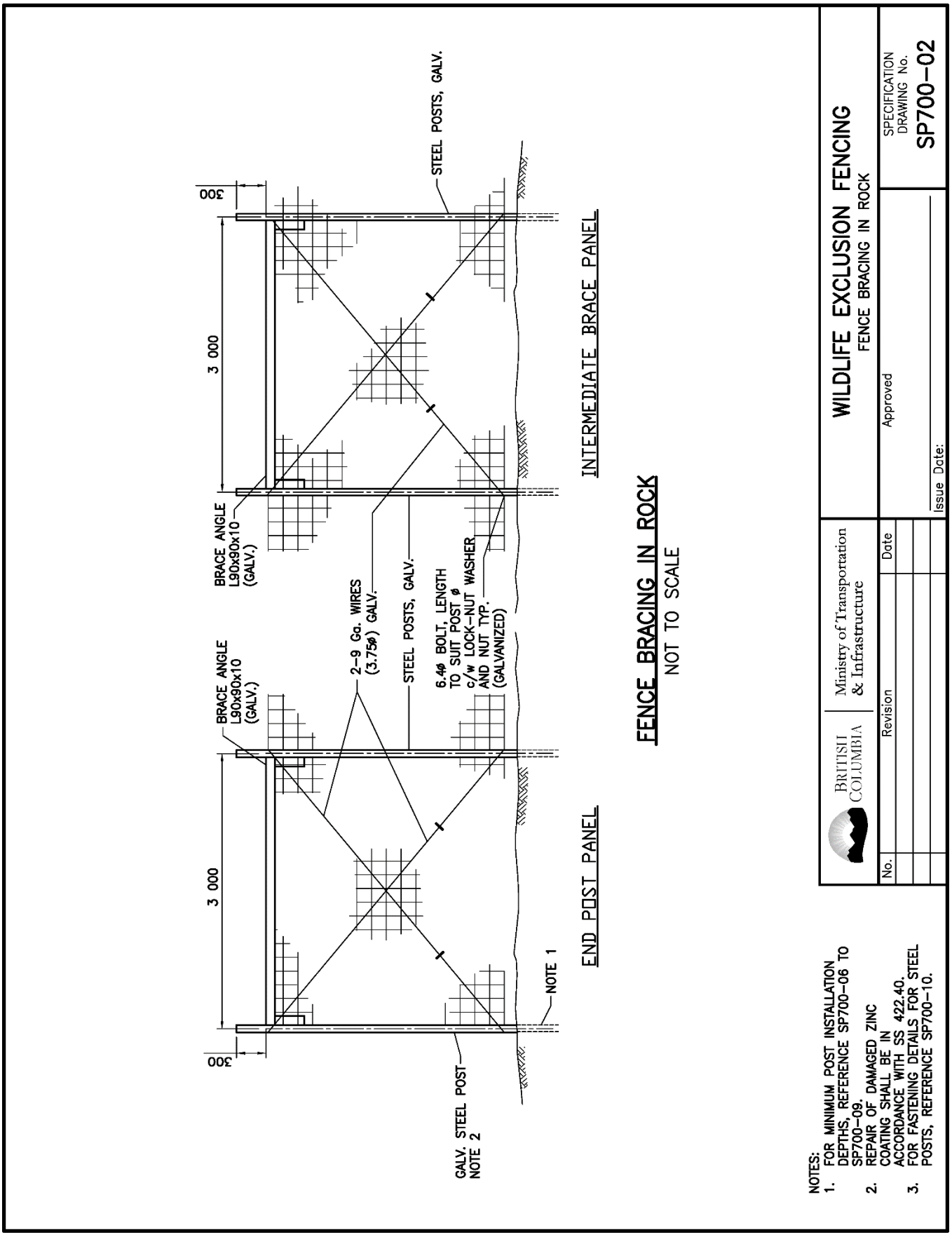
PAYMENT

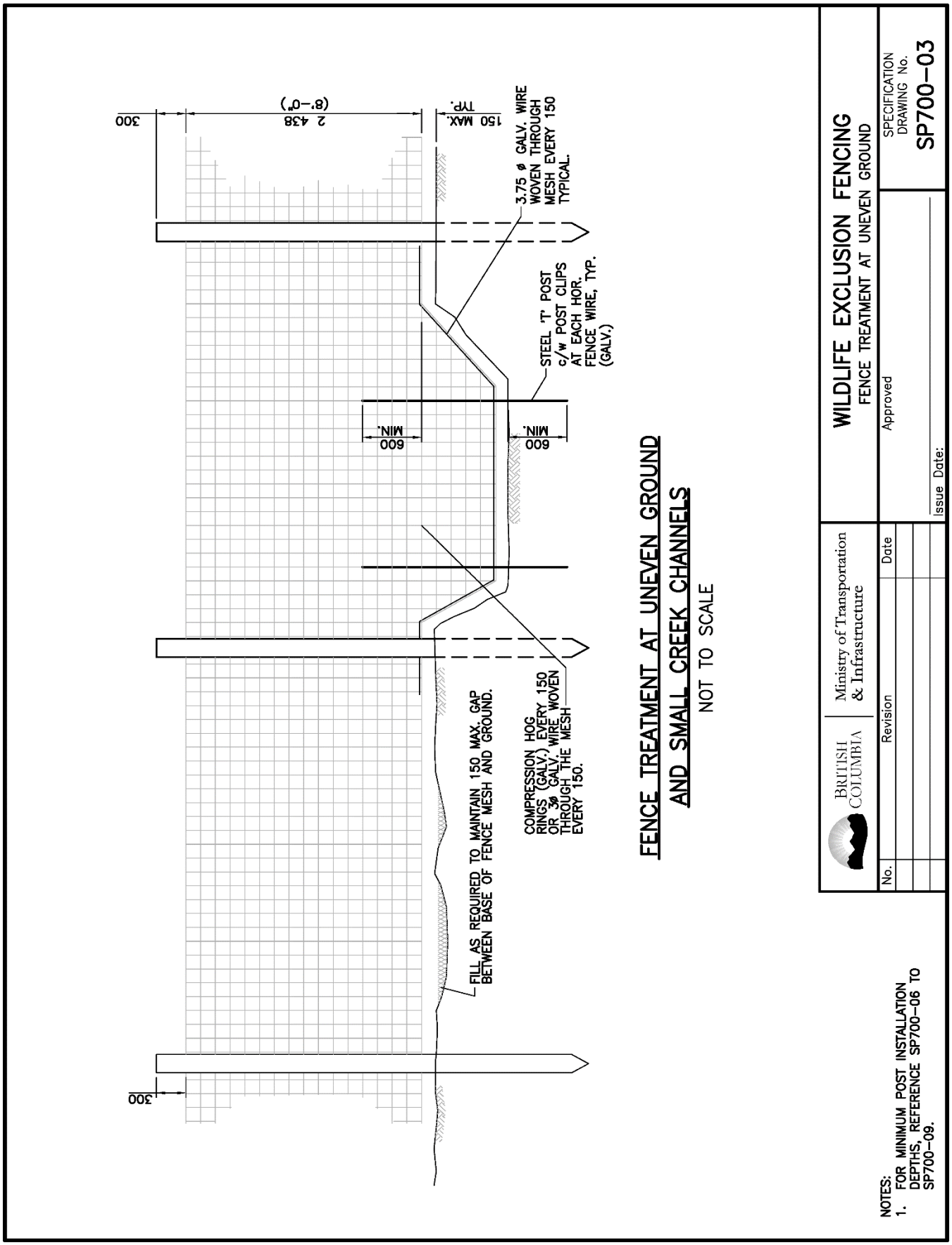
700.50 Fencing – Payment for FENCING, meeting the requirements as specified to the satisfaction of the Ministry Representative, will be at the Contract Unit Price per lineal metre of complete fencing, including any tensioning assemblies, but excluding one-way gates, lockable human access gates, double swing gates.

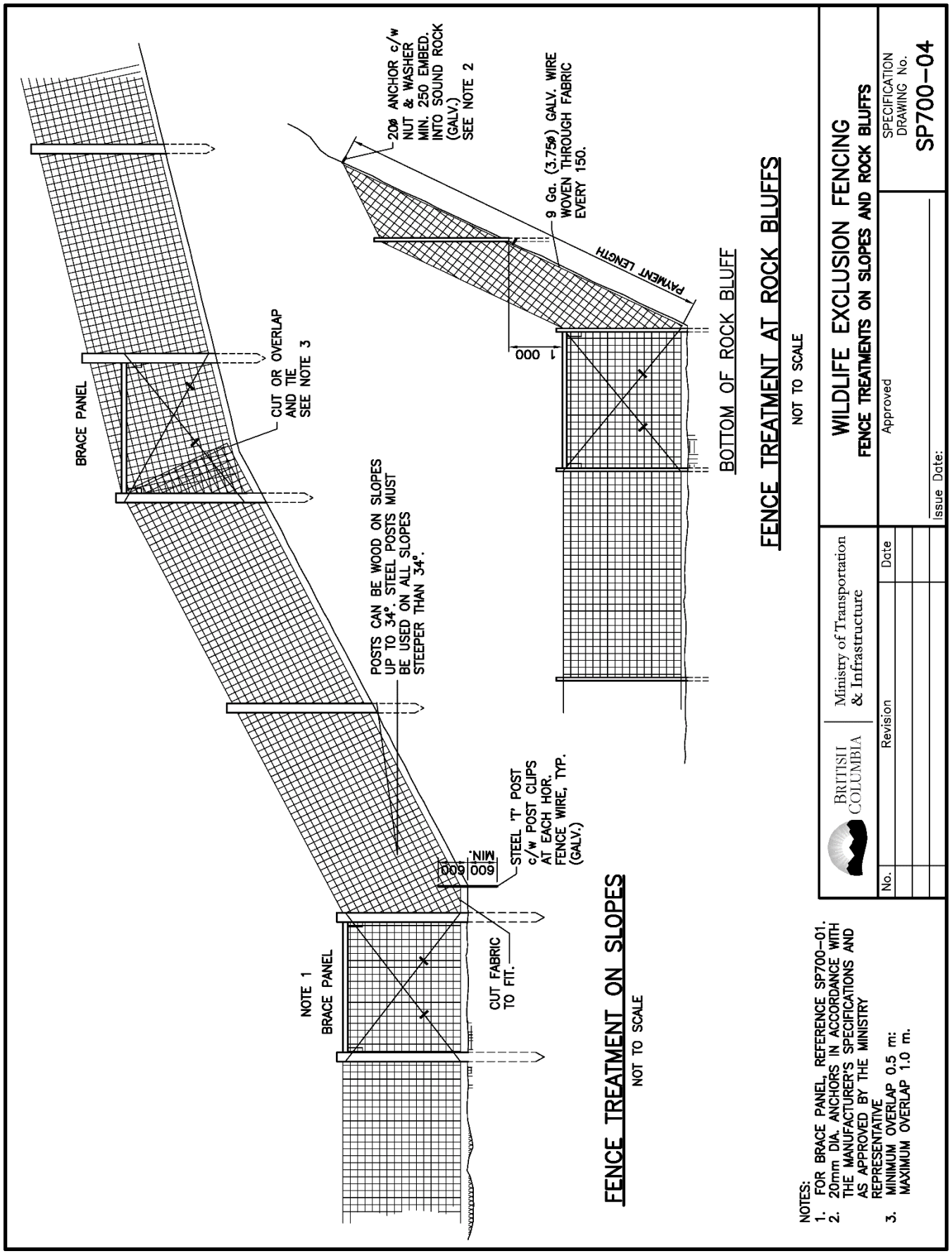
The Contract Unit Price(s) shall be accepted as full compensation for furnishing all material and/or taking delivery of Ministry supplied material as and where noted, all labour, tools, equipment and incidentals to complete the required installation, including the clearing of any additional right of way, construction of temporary fencing, connection to existing fences and/or structures, and final clean up.

700.51 Gates – Payment for ONE-WAY GATES, LOCKABLE HUMAN ACCESS GATES, and DOUBLE SWING GATES, meeting the requirements as specified to the satisfaction of the Ministry Representative, will be at the Contract Unit Price for each type and size furnished and/or installed complete in place.


FENCE BRACING (ALL SOILS EXCEPT ROCK)
NOT TO SCALE
END POST PANEL
INTERMEDIATE BRACE PANEL
100x100 TREATED WOOD COMPRESSION MEMBER
3 000
300
100x100x300 SUPPORT POST UNDER CROSS BRACE
250 LONG GALV. SPIKE PRE-DRILL HOLE TYP.
NOTE 1
NOTE 2
2-9 Ga. WIRES (3.75ø) GALV. WIRES
WOOD POST (TYP.)
ALL WOOD POSTS ARE 150 TO 200ø
100 LONG GALV. ARDOX NAIL TYP.
150
350
NOTES:
1. FOR MINIMUM POST INSTALLATION DEPTHS, REFERENCE SP700-06 TO SP700-09.
2. TENSION BATTEN LEFT IN PLACE.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
FENCE BRACING
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-01


STEEL POSTS, GALV.
300
3 000
BRACE ANGLE L90x90x10 (GALV.)
2-9 Ga. WIRES (3.75ø) GALV.
STEEL POSTS, GALV.
6.4ø BOLT, LENGTH TO SUIT POST ø c/w LOCK-NUT WASHER AND NUT TYP. (GALVANIZED)
INTERMEDIATE BRACE PANEL
BRACE ANGLE L90x90x10 (GALV.)
3 000
NOTE 1
END POST PANEL
300
GALV. STEEL POST NOTE 2
FENCE BRACING IN ROCK
NOT TO SCALE
NOTES:
1. FOR MINIMUM POST INSTALLATION DEPTHS, REFERENCE SP700-06 TO SP700-09.
2. REPAIR OF DAMAGED ZINC COATING SHALL BE IN ACCORDANCE WITH SS 422.40.
3. FOR FASTENING DETAILS FOR STEEL POSTS, REFERENCE SP700-10.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
WILDLIFE EXCLUSION FENCING
FENCE BRACING IN ROCK
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-02


FENCE TREATMENT AT UNEVEN GROUND
AND SMALL CREEK CHANNELS
NOT TO SCALE
3.75 ø GALV. WIRE WOVEN THROUGH MESH EVERY 150 TYPICAL.
STEEL 'T' POST c/w POST CLIPS AT EACH HOR. FENCE WIRE, TYP. (GALV.)
COMPRESSION HOG RINGS (GALV.) EVERY 150 OR 3ø GALV. WIRE WOVEN THROUGH THE MESH EVERY 150.
FILL AS REQUIRED TO MAINTAIN 150 MAX. GAP BETWEEN BASE OF FENCE MESH AND GROUND.
300
2 438 (8'-0")
150 MAX. TYP.
600 MIN.
600 MIN.
300
NOTES:
1. FOR MINIMUM POST INSTALLATION DEPTHS, REFERENCE SP700-06 TO SP700-09.
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No. Revision Date
WILDLIFE EXCLUSION FENCING
FENCE TREATMENT AT UNEVEN GROUND
Approved
Issue Date:
SPECIFICATION DRAWING No.
SP700-03


WILDLIFE EXCLUSION FENCING
FENCE TREATMENTS ON SLOPES AND ROCK BLUFFS
SPECIFICATION DRAWING No.
SP700-04
Approved
Issue Date:
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure
No.
Revision
Date
FENCE TREATMENT AT ROCK BLUFFS
NOT TO SCALE
20ø ANCHOR c/w NUT & WASHER MIN. 250 EMBED. INTO SOUND ROCK (GALV.) SEE NOTE 2
9 Ga. (3.75ø) GALV. WIRE WOVEN THROUGH FABRIC EVERY 150.
PAYMENT LENGTH
1 000
BOTTOM OF ROCK BLUFF
BRACE PANEL
CUT OR OVERLAP AND TIE SEE NOTE 3
POSTS CAN BE WOOD ON SLOPES UP TO 34°. STEEL POSTS MUST BE USED ON ALL SLOPES STEEPER THAN 34°.
600 MIN.
600
STEEL 'T' POST c/w POST CLIPS AT EACH HOR. FENCE WIRE, TYP. (GALV.)
NOTE 1
BRACE PANEL
CUT FABRIC TO FIT.
FENCE TREATMENT ON SLOPES
NOT TO SCALE
NOTES:
1. FOR BRACE PANEL, REFERENCE SP700-01.
2. 20mm DIA. ANCHORS IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS AND AS APPROVED BY THE MINISTRY REPRESENTATIVE.
3. MINIMUM OVERLAP 0.5 m:
MAXIMUM OVERLAP 1.0 m.